

Clostridioides difficile, a New "Superbug".

PMID: 37110267 | Indexed in PubMed

Clostridioides difficile is a Gram-positive, spore-forming, anaerobic bacterium. The clinical features of *C. difficile* infections (CDIs) can vary, ranging from the asymptomatic carriage and mild self-limiting diarrhoea to severe and sometimes fatal pseudomembranous colitis. *C. difficile* infections (CDIs) are associated with disruption of the gut microbiota caused by antimicrobial agents. The infections are predominantly hospital-acquired, but in the last decades, the CDI patterns have changed. Their prevalence increased, and the proportion of community-acquired CDIs has also increased. This can be associated with the appearance of hypervirulent epidemic isolates of ribotype 027. The COVID-19 pandemic and the associated antibiotic overuse could additionally change the patterns of infections. Treatment of CDIs is a challenge, with only three appropriate antibiotics for use. The wide distribution of *C. difficile* spores in hospital environments, chronic persistence in some individuals, especially children, and the recent detection of *C. difficile* in domestic pets can furthermore worsen the situation. "Superbugs" are microorganisms that are both highly virulent and resistant to antibiotics. The aim of this review article is to characterise *C. difficile* as a new member of the "superbug" family. Due to its worldwide spread, the lack of many treatment options and the high rates of both recurrence and mortality, *C. difficile* has emerged as a major concern for the healthcare system.

Gut Microbiota Composition Associated with *Clostridioides difficile* Colonization and Infection.

PMID: 35890026 | Indexed in PubMed

Clostridioides difficile is an anaerobic Gram-positive and spore-forming bacterium. The majority of *C. difficile* strains produce two toxins, A and B, associated with the development of acute diarrhea and/or colitis. In this review, two situations are distinguished: *C. difficile* infection (CDI) and asymptomatic colonization (AC). The main objective of this review is to explore the available data related to the link between the gut microbiota and the development of CDI. The secondary aim is to provide more information on why some people colonized with toxigenic *C. difficile* develop an infection while others show no signs of disease. Several factors, such as the use of antibiotics and proton pump inhibitors, hospitalization, and age, predispose individuals to *C. difficile* colonization and/or *C. difficile* infection. The gut microbiota of people with AC showed decreased abundances of *Prevotella*, *Alistipes*, *Bacteroides*, *Bifidobacterium*, *Dorea*, *Coprococcus*, and *Roseburia*. The gut microbiota of people suffering from CDI showed reductions in the abundances of *Lachnospiraceae*, *Ruminococcaceae*, *Blautia* spp., *Prevotella* spp., *Dialister* spp., *Bifidobacterium* spp., *Roseburia* spp., *Anaerostipes* spp., *Faecalibacterium* spp. and *Coprococcus* spp., in comparison with healthy people. Furthermore, increases in the abundances of *Enterococcaceae* and *Enterococcus* were associated with *C. difficile* infection.

Gut Microbiota Associated with *Clostridioides difficile* Carriage in Three Clinical Groups (Inflammatory Bowel Disease, *C. difficile* Infection and Healthcare Workers) in Hospital Field.

PMID: 37894185 | Indexed in PubMed

Clostridioides difficile is an anaerobic spore-forming Gram-positive bacterium. *C. difficile* carriage and 16S rDNA profiling were studied in three clinical groups at three different sampling times: inflammatory bowel disease (IBD) patients, *C. difficile* infection (CDI) patients and healthcare workers (HCWs). Diversity analysis was realized in the three clinical groups, the positive and negative *C. difficile* carriage groups and the three analysis periods. Concerning the three clinical groups, β -diversity tests showed significant differences between them, especially between the HCW group and IBD group and between IBD patients and CDI patients. The Simpson index (evenness) showed a significant difference between two clinical groups (HCWs and IBD). Several genera were significantly different in the IBD patient group (*Sutterella*, *Agathobacter*) and in the CDI patient group (*Enterococcus*, *Clostridioides*). Concerning the positive and negative *C. difficile* carriage groups, β -diversity tests showed significant differences. Shannon, Simpson and InvSimpson indexes showed significant differences between the two groups. Several genera had significantly different relative prevalences in the negative group (*Agathobacter*, *Sutterella*, *Anaerostipes*, *Oscillospira*) and the positive group (*Enterococcus*, *Enterobacteriaceae_ge* and *Enterobacterales_ge*). A microbiota footprint was detected in *C. difficile*-positive carriers. More experiments are needed to test this microbiota footprint to see its impact on *C. difficile* infection.

Introduction to the Special Issue on *Clostridioides difficile* Infection, Second Edition.

PMID: 39061289 | Indexed in PubMed

Clostridioides difficile (CD) is a Gram-positive, anaerobic bacterium that is one of the most common causes of infective diarrhoea worldwide [...].

A Three-Year Analysis of Mortality in *Clostridioides difficile* Patients in a Tertiary Center.

PMID: 39717327 | Indexed in PubMed

Clostridioides difficile, an anaerobic bacillus ubiquitous in nature, is the leading cause of hospital-acquired diarrhoea and one of the main causes of mortality by nosocomial infections. We aimed to identify the main predictors of the risk of dying and the characteristics of a three-year cohort of patients hospitalised in our clinic that eventually had an unfavourable outcome. We collected retrospectively available data for all patients hospitalised between January 1, 2021, and December 31, 2023. The characteristics of the patients who died after the CDI (*Clostridioides difficile* infection) were analysed and compared with those of the patients who survived. In the three-year interval mentioned above, 1086 patients had the main or secondary diagnosis of CDI. Of these, 97 patients (8.93%) died. The overall mortality for the same period was 2.62%. Eight patients (8.24%) who died had the primary diagnosis of CDI, while in the entire group, the percentage of patients with a primary diagnosis was 54.7%. Statistically significant differences between the groups of deceased and survivor patients were found for the following parameters: age ($p < 0.001$, 95% CI (confidence interval): 12.5-20.5), previous CDI episodes ($p = 0.033$, 95% CI: 0.014-0.329), and for the following parameters measured at admission: systolic blood pressure, quick sepsis-related organ failure assessment (qSOFA), leucocyte count, haemoglobin, creatinine, albumin, potassium, INR (international normalised ratio), CRP (C-reactive protein), fibrinogen, and procalcitonin. The number of hospitalisation days for the patients who died was significantly higher ($p < 0.001$, 95% CI: 4.3-12.6.). We identified the characteristics that significantly differentiated the patients who died from those who survived. Mortality is significantly higher in the group of patients with CDI than that in the other hospitalised patients.

Draft genome sequences of two distinct *Clostridioides difficile* isolates coinfecting a patient.

PMID: 39655932 | Indexed in PubMed

Clostridioides difficile is a spore-forming anaerobic Gram-positive bacillus. In this study, two distinct strains of *C. difficile* causing coinfection were isolated from the same fecal sample. Whole-genome sequencing was performed on these two strains to elucidate their genomic characteristics.

Review and Commentary on the Importance of Bile Acids in the Life Cycle of *Clostridioides difficile* in Children and Adults.

PMID: 33626138 | Indexed in PubMed

Clostridioides difficile, a spore-forming anaerobe, resides in the intestine. The life cycle of *C. difficile* illustrates an interdependent relationship between bile acids, commensal microbiota, and *C. difficile*. Primary bile acids are critical for the germination of *C. difficile* spores in the small intestine, while secondary bile acids serve as a counterbalance to inhibit the growth of the organism in the colon. Many commensal bacteria especially *Clostridium* spp. are responsible for transforming primary bile acids into secondary bile acids. Antibiotics eliminate bacteria that convert primary bile acids into secondary bile acids and, thus, allow *C. difficile* to flourish and cause diarrhea. In children younger than 2 years of age, who normally only produce primary bile acids, colonization with toxin-producing *C. difficile* is exceedingly common. The reason for the absence of *C. difficile* diarrhea in the children remains unexplained.

Clostridioides difficile Biofilm.

PMID: 38175479 | Indexed in PubMed

Clostridioides difficile infection (CDI), previously *Clostridium difficile* infection, is a symptomatic infection of the large intestine caused by the spore-forming anaerobic, gram-positive bacterium *Clostridioides difficile*. CDI is an important healthcare-associated disease worldwide, characterized by high levels of recurrence, morbidity, and mortality. CDI is observed at a higher rate in immunocompromised patients after antimicrobial therapy, with antibiotics disrupting the commensal microbiota and promoting *C. difficile* colonization of the gastrointestinal tract. A rise in clinical isolates resistant to multiple antibiotics and the reduced susceptibility to the most commonly used antibiotic molecules have made the treatment of CDI more complicated, allowing the persistence of *C. difficile* in the intestinal environment. Gut colonization and biofilm formation have been suggested to contribute to the pathogenesis and persistence of *C. difficile*. In fact, biofilm growth is considered as a serious threat because of the related antimicrobial tolerance that makes antibiotic therapy often ineffective. This is the reason why the involvement of *C. difficile* biofilm in the pathogenesis and recurrence of CDI is attracting more and more interest, and the mechanisms underlying biofilm formation of *C. difficile* as well as the role of biofilm in CDI are increasingly being studied by researchers in the field. Findings on *C. difficile* biofilm, possible implications in CDI pathogenesis and treatment, efficacy of currently available antibiotics in treating biofilm-forming *C. difficile* strains, and some antimicrobial alternatives under investigation will be

discussed here.

The C-Terminal Domain of *Clostridioides difficile* TcdC Is Exposed on the Bacterial Cell Surface.

PMID: 32868401 | Indexed in PubMed

Clostridioides difficile is an anaerobic Gram-positive bacterium that can produce the large clostridial toxins toxin A and toxin B, encoded within the pathogenicity locus (PaLoc). The PaLoc also encodes the sigma factor TcdR, which positively regulates toxin gene expression, and TcdC, which is a putative negative regulator of toxin expression. TcdC is proposed to be an anti-sigma factor; however, several studies failed to show an association between the *tcdC* genotype and toxin production. Consequently, the TcdC function is not yet fully understood. Previous studies have characterized TcdC as a membrane-associated protein with the ability to bind G-quadruplex structures. The binding to the DNA secondary structures is mediated through the oligonucleotide/oligosaccharide binding fold (OB-fold) domain present at the C terminus of the protein. This domain was previously also proposed to be responsible for the inhibitory effect on toxin gene expression, implicating a cytoplasmic localization of the OB-fold. In this study, we aimed to obtain topological information on the C terminus of TcdC and demonstrate that the C terminus of TcdC is located extracellularly. In addition, we show that the membrane association of TcdC is dependent on a membrane-proximal cysteine residue and that mutating this residue results in the release of TcdC from the bacterial cell. The extracellular location of TcdC is not compatible with the direct binding of the OB-fold domain to intracellular nucleic acid or protein targets and suggests a mechanism of action that is different from that of the characterized anti-sigma factors.

IMPORTANCE The transcription of *C. difficile* toxins TcdA and TcdB is directed by the sigma factor TcdR. TcdC has been proposed to be an anti-sigma factor. The activity of TcdC has been mapped to its C terminus, and the N terminus serves as the membrane anchor. Acting as an anti-sigma factor requires a cytoplasmic localization of the C terminus of TcdC. Using cysteine accessibility analysis and a HiBiT-based system, we show that the TcdC C terminus is located extracellularly, which is incompatible with its role as anti-sigma factor. Furthermore, mutating a cysteine residue at position 51 resulted in the release of TcdC from the bacteria. The codon-optimized version of the HiBiT (HiBiTopt) extracellular detection system is a valuable tool for topology determination of membrane proteins, increasing the range of systems available to tackle important aspects of *C. difficile* development.

Culturing and maintaining *Clostridium difficile* in an anaerobic environment.

PMID: 24084491 | Indexed in PubMed

Clostridium difficile is a Gram-positive, anaerobic, sporogenic bacterium that is primarily responsible for antibiotic associated diarrhea (AAD) and is a significant nosocomial pathogen. *C. difficile* is notoriously difficult to isolate and cultivate and is extremely sensitive to even low levels of oxygen in the environment. Here, methods for isolating *C. difficile* from fecal samples and subsequently culturing *C. difficile* for preparation of glycerol stocks for long-term storage are presented. Techniques for preparing and enumerating spore stocks in the laboratory for a variety of downstream applications including microscopy and animal studies are also described. These techniques necessitate an anaerobic chamber, which maintains a consistent anaerobic environment to ensure proper conditions for optimal *C. difficile*

growth. We provide protocols for transferring materials in and out of the chamber without causing significant oxygen contamination along with suggestions for regular maintenance required to sustain the appropriate anaerobic environment for efficient and consistent *C. difficile* cultivation.
